Security risk assessment report

Part 1: Select up to three hardening tools and methods to implement
For the 1st and the 2nd vulnerabilities: The organization’s employees' share passwords // The admin password for the database is set to the default. Hardening tool to implement : Password policies To put password policies focuses on using methods to salt and hash passwords, rather than requiring overly complex passwords or enforcing frequent changes to passwords. For the 3rd vulnerability : The firewalls do not have rules in place to filter traffic coming in and out of the network. Hardening tools to implement 1- Disabling unused ports 2- Port filtering 3- Firewall maintenance For the 4th vulnerability : Multifactor authentication (MFA) is not used. Hardening tools to implement 1- Multifactor authentication (MFA)

Part 2: Explain your recommendations
Password Policies & Credential Management Why it is effective: Implementing strong password policies and secure backend practices (like salting and hashing) prevents attackers from easily guessing credentials, either manually or through automated scripts executing brute-force attacks. It also eliminates the vulnerability of default administrative credentials. Implementation/Frequency: Enforced continuously via system configurations and updated periodically based on evolving threat intelligence. Disabling Unused Ports Why it is effective: This practice closes unnecessary network pathways, preventing malicious actors from exploiting open, unmonitored ports to infiltrate the network. It can also be applied as an immediate corrective measure following a security incident. Implementation/Frequency: Performed regularly during system audits and network configuration reviews. Port Filtering Why it is effective: Port filtering acts as a gatekeeper to control incoming and outcoming network traffic based on predefined security rules, preventing unauthorized users or malicious payloads from entering the private network. Implementation/Frequency: Implemented continuously and monitored in real time by network security controls. Firewall Maintenance Why it is effective: Regularly maintaining and updating firewall rules ensures that abnormal or malicious traffic is successfully blocked. Timely rule updates help protect the network against various threats, including Distributed Denial of Service (DDoS) attacks and unauthorized lateral movement. Implementation/Frequency: Performed routinely (e.g., monthly or quarterly) and updated immediately following any major infrastructure changes or security incidents. Multifactor Authentication (MFA) Why it is effective: MFA adds an extra layer of security beyond just a username and password. Even if an attacker manages to obtain or brute-force a user's

password, they cannot gain access without the second verification factor (such as an authenticator app code or hardware token).

Implementation/Frequency: Configured once and maintained continuously across all user and administrative accounts.